

Make sure to show all work to receive full credit

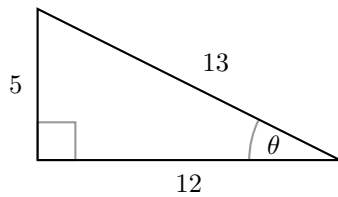
Score: _____ out of 43

Important: Leave all answers as an exact value unless otherwise stated.

1. Given the triangle below, evaluate

(a) (2 points) $\tan \theta$.

(b) (2 points) $\sec \theta$.



2. (5 points) Joule (my other cat) is flying a kite. The kite is extended horizontally 35 feet and flying at an angle of 30° to the ground. How high above the ground is the kite?

3. (5 points) If $\sin \theta = -\frac{\sqrt{2}}{5}$ and θ lies in quadrant III, find the exact value of $\sec \theta$.

4. (3 points) If possible, find the exact value of $\csc^{-1}(2)$.

5. (3 points) Find the exact value of $\sin\left(\frac{\pi}{3}\right)$.

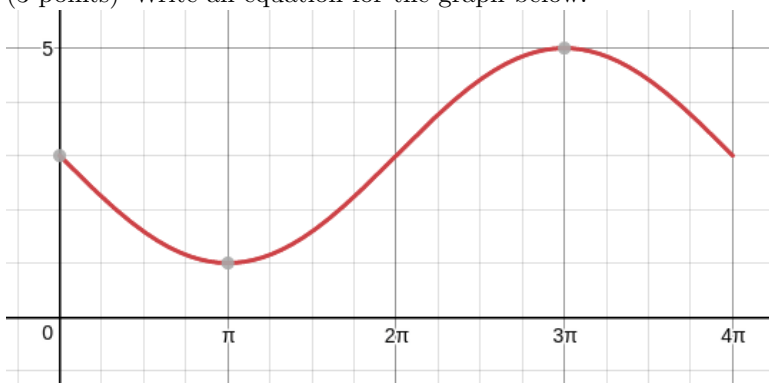
6. (5 points) Verify the identity

$$(\tan^2 x + 1)(\cos^2 x - 1) = -\tan^2 x$$

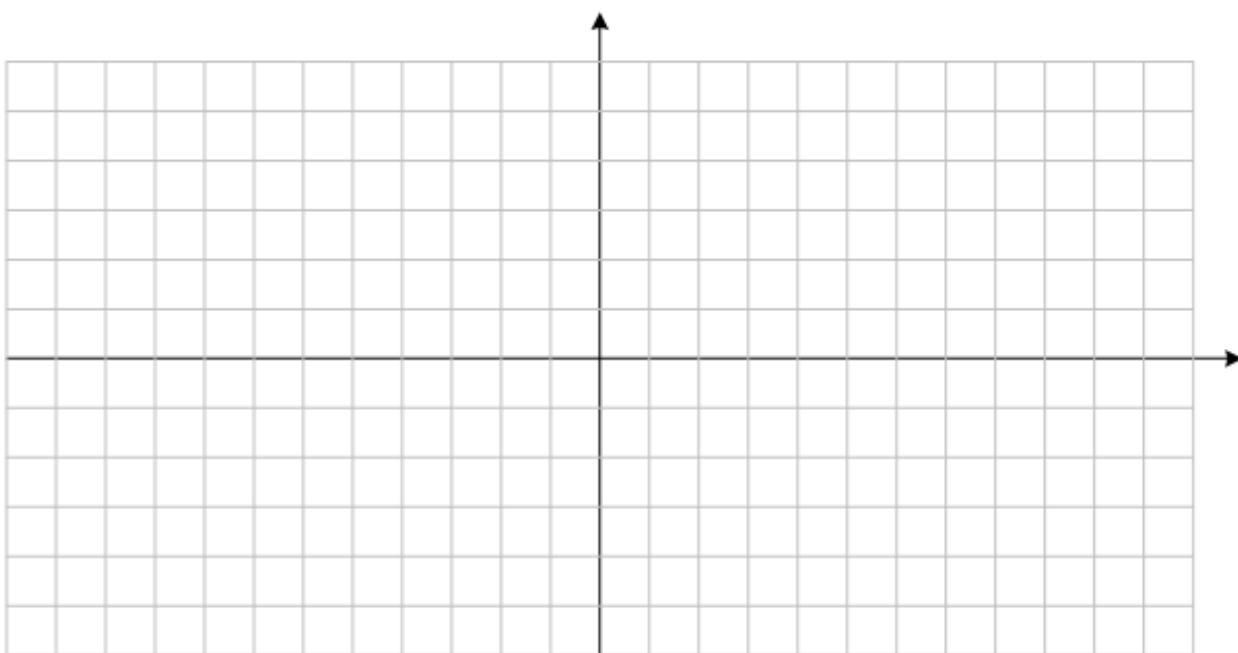
7. (5 points) Find ALL the exact solutions of $\sqrt{3}\sec(x) + 2 = 0$

8. (5 points) If possible, write $\tan(\sin^{-1}(\frac{2}{x}))$ as an algebraic expression of x . Hint: Use a right triangle.

9. (3 points) Write an equation for the graph below.



10. (5 points) Graph one period of $y = -3\cos\left(\frac{\pi}{6}(x - 2)\right)$ in its “Normal Viewing Window” (Just like in class). Draw at least one intermediate step (or graph) between your initial graph and final answer.



Formulas

